MTH 336(A) Dr. James Ashe
Exam 1 Friday 20$^{\text{th}}$ September, 2013 **Name:**

Show your work.

1. [3 pts each] Let $\mathbf{x} = (1, 2, 2)$ and $\mathbf{y} = (3, 0, 2)$.

a. Find $\mathbf{x} - \mathbf{y}$.

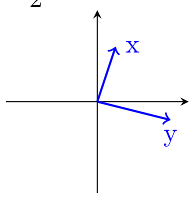
b. Find $\mathbf{x} \cdot \mathbf{y}$.

c. Find $\|\mathbf{y}\|$.

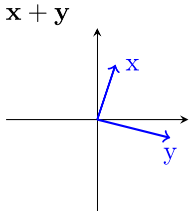
d. Find a unit vector in the direction of y.

2. [3 pts each] Use the provided graph to sketch the following vectors.

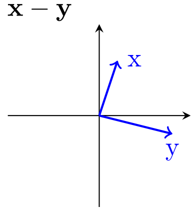
a. $-\frac{1}{2}\mathbf{y}$



b. $\mathbf{x} + \mathbf{y}$



c. $\mathbf{x} - \mathbf{y}$



3. [5 pts each] Give Cartesian equations of the given hyperplanes.

a. The hyperplane in $\mathbb{R}^2$ passing through $(1, 1)$ and orthogonal to the line $\mathbf{x} = (0, 2) + t(2, 1)$

b. The hyperplane in $\mathbb{R}^3$ which contains the point $(0, 1, 0)$ and has normal vector $(1, 2, 1)$.

4. Consider the matrix,

$$A = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 2 & 0 & 0 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

a. [4 pts] Find the reduced echelon form of A.

b. [4pts] Give the general solution of $A\mathbf{x} = \mathbf{0}$ in *standard* form.

c. [2 pts] Is the reduced echelon form of any matrix unique? i.e., if two different people compute the reduced echelon form of the same matrix, will they always arrive at the exact same result?

5. Consider the system of equations

$$\begin{array}{rrrl} x_1 & & +x_3 & = 3 \\ & x_2 & +2x_3 & = 1 \\ & 2x_2 & +4x_3 & = 2 \end{array}$$

- **a.** [2 pts] Express the system as an augmented matrix.
- **b.** [4 pts] Find the reduced echelon form of this augmented matrix.
- **c.** [4 pts] Give the general solution to the system of equations.

6. [3 pts each] Consider the matrices

$$A = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

a. How many solutions does the system of equations represented by $A\mathbf{x} = \mathbf{b}$ have?

b. What is the *rank* of A?

c. How many constraints does the solutions of $A\mathbf{x} = \mathbf{b_1}$ have if

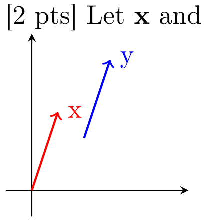
$$\mathbf{b_1} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

7. Complete the following

a. [2 pts] Suppose that the vectors $\mathbf{v_1}$, $\mathbf{v_2}$, and $\mathbf{v_3}$ span the hyperplane $\mathcal{H}$, and $\mathbf{w} = 2\mathbf{v_1} - \mathbf{v_2} + 3\mathbf{v_3}$. Is $\mathbf{w}$ in $\mathcal{H}$?

b. [2 pts] Suppose that the $n \times n$ matrix A is nonsingular. Does $A\mathbf{x} = \mathbf{b}$ have a solution? If so, is it unique? (Recall that nonsingular means that the reduced echelon form of the matrix does not have a row with all zeros.)

c. [2 pts] Let $\mathbf{x}$ and $\mathbf{y}$ be vectors shown in the figure below. Does $\mathbf{x} = \mathbf{y}$?



d. [4 pts] Let

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}.$$

Find $A + B$.

8. [Bonus] Prove that $n \times n$ matrices have an identity element under matrix addition and multiplication, i.e., that given any $n \times n$ matrix A there exists a matrix B and C such that $A + B = A$ and $AC = A$.